DIGITAL SAT & EST MATH COURSES
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ESt1 • System of Equations — Lesson

Solving Systems of Linear & Nonlinear Equations

Student Name: _____ **Date:** _____ **Score:** / 23

1 Fundamental Principles of Systems of Equations

A **system of equations** is two (or more) equations solved **together**. The solution (x, y) is the point where the graphs meet — it must satisfy **every** equation at the same time. The graphs tell you how many solutions the system has:

One solution
The lines **intersect at a single point** (different slopes).

No solution
The lines are **parallel** (same slope, different intercepts).

Infinitely many
The lines are **confounded** — they are the same line.

Quick coefficient test for $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$:

Type of solution	Condition on the coefficients
One solution	$\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$
No solution	$\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$
Infinitely many	$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

Three ways to find the solution:

Substitution
Replace one variable using the other equation (e.g. $y = 4x$), then solve.

Elimination
Multiply an equation so one variable cancels, then add or subtract the equations.

Calculator
Casio equation mode (**Mode 5 1**): type both equations and read x and y.

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Eng Rabaa Farrag Way

THE EASY WAY

A Linear system (two lines) — use the calculator

1. **Rearrange** the equations into the form $ax + by = c$ (x and y on the left, the constant on the right). An equation that is already in this form does not need rearranging.
2. On your calculator press MODE then 5, then 1.
3. **Enter** the coefficient of *x*, the coefficient of *y* and the constant of each equation — you get the values of *x* and *y*.

Note: if your calculator is model **fx-991ARX**, press MENU, then 9, then 2, then 1.

B Linear + nonlinear system (a line and a parabola) — use Shift Solve

1. **Set the two expressions equal** (for example $x^2 - x - 20 = 2x + 3$) and use SHIFT SOLVE = to find the **first value of *x***.
2. Use SHIFT SOLVE with **100** (or **-100**) as the starting value to find the **second value of *x***.
3. **Substitute** each value of *x* in the first equation to find *y*.

Why it is powerful: it turns long algebra into a few key presses, saves time on the exam, and lets you **check your answer instantly** by putting *x* and *y* back into the equations.

2 Solved Examples (ESt1 Style) — Eng Rabaa Farrag Way

Example 1 — Linear system (calculator)

Find the solution of this system of linear equations:

$$\begin{cases} 2x = -y + 7 \\ x - y = -1 \end{cases}$$

1. **Rearrange the first equation** so that *y* moves to the left side:

$$2x + y = 7$$

The second equation, $x - y = -1$, does **not** need rearranging because it is already in the form $ax + by = c$.

2. On your calculator press MODE then 5, then 1. (fx-991ARX: MENU 9 2 1)
3. Enter the coefficients of *x* and *y* and the constant of each equation:

x	y	constant
2	1	7
1	−1	−1

4. Read the values of *x* and *y* from the calculator:

$$x = 2 \quad , \quad y = 3$$

✓ Check: 2(2) = -3 + 7 = 4 and 2 - 3 = -1

Example 2 — A line and a parabola (nonlinear system)

Find the solution of this system:

$$\begin{cases} y = 2x + 3 \\ y = x^2 - x - 20 \end{cases}$$

1 **Set the two expressions equal** and use SHIFT SOLVE = to find the **first value of *x***:

$$x^2 - x - 20 = 2x + 3$$

2 Use SHIFT SOLVE with **100** (or **-100**) as the starting value to find the **second value of *x***:

START 100	START -100
$x \approx 6.52$	$x \approx -3.52$

3 **Substitute** each value of *x* in the first equation, $y = 2x + 3$, to find *y*:

x ≈ 6.52	x ≈ -3.52
$2(6.52) + 3 \approx 16.05$	$2(-3.52) + 3 \approx -4.05$

4 The system has **two solutions** (the line cuts the parabola at two points):

$$(x,\ y) \approx (6.52,\ 16.05) \quad \text{and} \quad (-3.52,\ -4.05)$$

3 Practice Questions

Choose the **one** correct answer for each question — 23 questions in total. Try the **Eng Rabaa Farrag Way** on the **calc** questions, and solve the **no calc** ones by hand.

Q1

ESt1 • 2021 M10 • no calc • Q2

Amina went to the flower shop and bought 2 roses and 5 daisies for 6 EGP. Lara bought from the same shop, 4 roses and 2 daisies for 4 EGP. How much should Ahmad pay to buy 2 roses and 2 daisies?

A 1 EGP

B 1.5 EGP

C 2 EGP

D 3 EGP

Q2

ESt1 • 2020 M12 • no calc • Q8

$$\begin{cases} 3y = 12 - 3y \\ y + a = x - 1 \end{cases}$$

In the system of equations above, a is a constant and (x, y) is a solution, where $x = 3$. What is the value of a?

A -4
B 0
C 2
D 4

Q3

ESt1 • 2021 M5 • no calc • Q1

If $2x - 3y = 14$ and $-x + 2y = -8$, what is the value of $x + y$?

A -2
B 0
C 2
D 6

Q4

ESt1 • 2022 M6 • no calc • Q1

Given $2x + y = 8$ and $3y + x = -1$. What is the value of $y - 2x$?

A -16
B -12
C -8
D 8

Q5

ESt1 • 2022 M3 • calc • Q3

The price of three squash and two zucchini is 10 EGP, while the price of five squash and three zucchini is 16.5 EGP. What is the price of one squash and four zucchini?

A 7 EGP
B 5 EGP
C 10 EGP
D 8.5 EGP

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Q6

ESt1 • 2022 M6 • calc • Q18

Shahira and Nicole went to the supermarket to buy chocolate. The list of what they bought and the total price paid can be seen in the table below.

	Number of chocolates bought from brand "A"	Number of chocolates bought from brand "B"	Total price paid (in $)
Shahira	7	3	25.4
Nicole	4	5	24.7

If you went to the same supermarket and bought 1 chocolate from brand "A" and 3 chocolates from brand "B", how much will you pay?

A $\$5.4$

B $\$7.5$

C $\$10$

D $\$11.6$

Q7

ESt1 • 2020 M12 • calc • Q27

$$\begin{cases} dx - 2my = -14 \\ 3dx + 6my = 78 \end{cases}$$

(d and m are constants). Given that the solution of the system above is (2, 5), what is the value of $2mx - 3dy$?

A -37

B -18

C 11

D 20

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Q8

ESt1 • 2021 M3 • calc • Q4

$$\begin{cases} \frac{2}{3}x + y = -3 \\ \frac{x}{3} + \frac{y}{2} = -3 \end{cases}$$

What is the solution of the above system?

A $(-6, 1)$

B $(6, 9)$

C $(6, -7)$

D $\varnothing$

Q9

ESt1 • 2021 M5 • calc • Q1

Which of the following is a point of intersection between the line with equation $y = 3x + 9$ and the parabola with the equation $y = 2x^2 + 3x + 1$?

A $(2, 3)$

B $(2, 15)$

C $(-2, 15)$

D $(-2, 2)$

Q10

ESt1 • 2021 M5 • calc • Q3

$$\begin{cases} -2x + 5y = 39 \\ 3x = -4y + 45 \end{cases}$$

From the system of equations above, what is the value of $2x + 7y$?

A -57

B 15

C 39

D 69

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Q11

ESt1 • 2022 M3 • calc • Q1

Which of the following is a point of intersection between the line with the equation $y = 3x + 2$ and the parabola with the equation $y = 4x^2 - 9x + 11$?

A $(3.5, 3)$

B $(4, 14)$

C $(-2, -4)$

D $(1.5, 6.5)$

Q12

ESt1 • 2020 M12 • no calc • Q16

$$\begin{cases} 13x - 7y = 12 \\ 7x - 13y = 6 \end{cases}$$

Based on the system of equations above, what is the value of $4x + 4y$?

A 1

B 2

C 3

D 4

Q13

ESt1 • 2021 M8 • no calc • Q5

A 100-page album costs twice as much as a 50-page album. The cost of three 100-page albums and two 50-page albums is $\$t$. How much does a 50-page album cost?

A $8t$

B $4t$

C $\frac{t}{4}$

D $\frac{t}{8}$

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Q14

ESt1 • 2021 M8 • no calc • Q4

A truck contains 15 identical boxes that are either red or blue.
The red box weighs 3 kg and the blue box weighs 2 kg.
If the total weight of the boxes is 36 kgs, what is the difference between the red and blue boxes in the truck?

A. 6
B. 9
C. 1
D. 3

Q15

ESt1 • 2022 M3 • no calc • Q1

If $x + y = 500$ and $3x + 2y = 3000$, what is the value of $5x - y$?

A. 8000
B. 7000
C. 11500
D. -7000

Q16

EST • February 2022 • no calc • Q4

When a system of two linear equations has a unique solution, how do the graphs of the equations appear?

A. The lines are parallel.
B. The lines are confounded.
C. The lines intersect at a single point.
D. None of the above

Q17

ESt1 • 2021 M3 • calc • Q5

When a system of two linear equations has no solution, how do the graphs of the equations appear?

A. The lines intersect at a single point.
B. The lines have the same x-intercept.
C. The lines are parallel.
D. The lines are confounded.

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Q18

ESt1 • 2020 M12 • no calc • Q10

$$\begin{cases} ax - \frac{1}{2}y = c \\ 2x + 4y = 5 \end{cases}$$

The system of equations above has infinitely many solutions. If a and c are constants, what is the value of c?

A $-\frac{5}{8}$

B $\frac{1}{4}$

C $\frac{5}{4}$

D $\frac{11}{4}$

Q19

EST • October 2021 • calc • Q19

Which of the following systems has infinite number of solutions?

A $\begin{cases} 2x - 3y = 1 \\ x + 3y = 4 \end{cases}$

B $\begin{cases} -2x + y = -1 \\ 5y = -2x \end{cases}$

C $\begin{cases} 3x + y = 2 \\ x = -5y \end{cases}$

D $\begin{cases} -2x - 2y = 4 \\ y = -x - 2 \end{cases}$

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Q20

ESt1 • Dr John EX1 • no calc • Q9

$$\begin{cases} 2x + by = 10 \\ ax + 4y = 15 \end{cases}$$

In the system of equations above, a and b are constants and $a = 2b$. If the system has no solution, which of the following could be a possible value of a?

A -2

B $\frac{1}{2}$

C 4

D 8

Q21

ESt1 • Dr John EX1 • calc • Q21

For O.K theater tickets, a ticket for an adult is 5 dollars more than a ticket for a child. If a group of 6 adults and 10 children pay a total of 142 dollars, what is the cost, in dollars, of a ticket for one adult and one child?

A 19

B 18

C 17

D 16

Q22

ESt1 • Dr John EX2 • calc • Q24

$$\begin{cases} y = -\frac{1}{10}x^2 + k \\ y = 5 \end{cases}$$

In the system of equations above, k is a constant. For which of the following values of k does the system of equations have no real solution?

A 10

B 8.5

C 5

D -0.05

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Q23

ESt1 • Dr John EX2 • calc • Q26

$$\begin{cases} x - y = 5 \\ xy = 10 \end{cases}$$

In the equations above, what is the value of $x^2 + y^2$?

A 15

B 25

C 36

D 45

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4 Answer Key

Check your answers after you finish. For each one, put *x* and *y* back into the equations to make sure they work.

Q1 D	Q2 B	Q3 C	Q4 B
Q5 B	Q6 D	Q7 A	Q8 D
Q9 B	Q10 D	Q11 D	Q12 D
Q13 D	Q14 D	Q15 C	Q16 C
Q17 C	Q18 A	Q19 D	Q20 C
Q21 A	Q22 D	Q23 D	

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